

Q 13 : Arduino Pin Types

Pin Type	Explanation	Arduino UNO Example	Real IoT Use Case
Digital Input	Reads only two states: HIGH (1) or LOW (0). Used to receive signals from switches or sensors.	Push Button on Pin 2	Detecting whether a door is open or closed in a smart home.
Digital Output	Sends HIGH or LOW signals to control external devices.	LED on Pin 13	Turning a relay ON/OFF to control lights remotely.
Analog Input	Reads analog voltages (0-5V) and converts them to digital values (0-1023).	Potentiometer on A0	Reading temperature or soil moisture sensor values for smart farming.
PWM Output	Produces Pulse Width Modulation signals to simulate analog output. Available on pins 3, 5, 6, 9, 10, 11.	LED brightness control on Pin 9	Controlling the speed of a DC motor or adjusting LED brightness in smart lighting.
I2C / SPI Pins	Communication pins used to connect sensors and modules. I2C: SDA (A4), SCL (A5). SPI: D10-D13.	OLED Display or MPU6050 sensor	Communicating with multiple sensors in a weather monitoring or smart home system.